BIRLA INSTITUTE OF TECHNOLOGY & PILANI

Work Integrated Learning Programmes Division

AIMLCZG530 — NATURAL LANGUAGE PROCESSING

GROUP ASSIGNMENT: RESEARCH-ORIENTED NLP MINI PROJECT

Total weight	Duration	Group size	Submission
20 marks	30 days	4 students	One submission per group

1. Assignment purpose and scope

Each group will solve one clearly defined NLP research problem. The project must use a documented dataset or documented synthetic data, build a baseline and an advanced or comparison method, conduct a controlled evaluation, perform an ablation, analyse errors, and communicate the findings in a research-paper format. Students are not expected to train large language models from scratch, use paid APIs, collect private data, or build a commercial system.

A dataset is the material on which the problem is studied; it is not the problem itself. A complete project must build the required systems, compare their results, perform the required analysis, and answer the research question. Merely downloading a dataset, running a library model, and reporting one score is insufficient.

Students may use obtainable public datasets, public-domain or openly licensed documents, and carefully designed synthetic data wherever it supports the assigned problem. Synthetic-data generation, limitations, and possible train-test leakage must be documented.

2. Group allocation

Problem statement	Allocated groups	Distinctness requirement
PS-1: Statistical and neural language modelling	1, 6, 11	Different corpus genres, dataset variants, or synthetic-data designs.
PS-2: POS tagging under domain shift	2, 7, 12	Different source/target domains, datasets, or unknown-word strategies.
PS-3: Parsing and information extraction	3, 8, 13	Different sentence domains, grammar coverage, relation schemas, or synthetic constructions.
PS-4: Static versus contextual embeddings	4, 9, 14	Different classification datasets, tasks, or synthetic-data settings.
PS-5: Retrieval-augmented question answering	5, 10, 15	Different public collections, synthetic knowledge bases, or question sets.

3. Common requirements

- Use Python and a reproducible environment.
- Implement at least one baseline and one advanced or comparison method.
- Use a fixed seed where relevant and a held-out test set.
- Complete at least one ablation or controlled comparison.
- Report quantitative results and qualitative errors or failures, not only successful examples.
- Use obtainable public data or synthetic data generated from a documented procedure.
- Declare pretrained models, external APIs, borrowed code, and generative-AI assistance.

PS-1. Statistical and Neural Language Modelling

Allocated groups: 1, 6, 11

Problem: A language model predicts the next word. N-gram models often assign zero probability to rare or unseen sequences, while smoothing and neural models may generalise better.

Objective: Determine whether smoothing and a small neural language model improve prediction of sparse and unseen word sequences.

What the group must do:

- Use NLTK Brown Corpus, WikiText-2, or a documented synthetic corpus generated from grammars, templates, or topic mixtures. Sources: https://www.nltk.org/data.html and https://nvidia.github.io/OpenSeq2Seq/html/language-model.html.
- Build an unsmoothed bigram, trigram, smoothed or backoff n-gram, and small neural language model.
- Evaluate perplexity, vocabulary size, OOV rate, and runtime on the same test set.
- Analyse at least ten rare or unseen sequences and perform one ablation, such as smoothing versus no smoothing.
- Conclude which approach handles sparse data best and explain its failures.

PS-2. POS Tagging Under Domain Shift

Allocated groups: 2, 7, 12

Problem: A POS tagger trained on one type of text may perform poorly on another type, especially with unfamiliar words, spelling, abbreviations, and informal language.

Objective: Measure how POS-tagging performance changes across domains and identify which unknown-word strategy improves robustness.

What the group must do:

- Use UD English-EWT as a source domain, UD English-GUM or a separated EWT genre as a target domain, or documented synthetic tagged sentences. Sources: https://universaldependencies.org/treebanks/en_ewt/index.html and https://universaldependencies.org/treebanks/en_gum/index.html.
- Build a simple baseline, first-order HMM with Viterbi decoding, unknown-word handling, and a pretrained spaCy or Stanza comparator.

- Report accuracy, macro-F1, selected per-tag results, confusion matrix, and runtime.
- Analyse at least 25 errors by category and perform one ablation by removing unknown-word handling, suffix features, or smoothing.
- Conclude which tagger and unknown-word strategy is most robust under domain shift.

PS-3. Parsing and Information Extraction

Allocated groups: 3, 8, 13

Problem: A sentence can have more than one structure. Different constituency and dependency representations may resolve ambiguity and extract relations differently.

Objective: Compare constituency-style and dependency-style representations for structural ambiguity and simple relation extraction.

What the group must do:

- Use UD English-EWT or GUM, SemEval-2010 Task 8, public-domain or openly licensed sentences, or synthetic CFG/PCFG sentences with gold trees and relations. Sources: https://universaldependencies.org/en/ and https://aclanthology.org/S10-1006/.
- Construct a restricted CFG or PCFG, implement CKY, show two charts, reconstruct a parse, and implement a constrained arc-standard dependency simulation or parser.
- Use at least four ambiguity examples and extract at least 30 subject–verb–object or entity–relation–entity triples.
- Report manual verification and precision, recall, and F1 where applicable; perform one ablation such as removing PCFG probabilities.
- Conclude which representation is more useful for the selected ambiguity and extraction task.

PS-4. Static versus Contextual Embeddings for Text Classification

Allocated groups: 4, 9, 14

Problem: Text classifiers use different representations. TF-IDF and static embeddings provide limited context, while contextual models represent words according to surrounding text.

Objective: Determine whether contextual embeddings improve classification enough to justify their additional computation and reduced interpretability.

What the group must do:

- Use AG News, IMDb, 20 Newsgroups, BBC News, or a documented synthetic labelled corpus. Sources: https://huggingface.co/datasets/ag_news, https://huggingface.co/datasets/stanfordnlp/imdb, and https://scikit-learn.org/stable/datasets/real_world.html.
- Build a TF-IDF classifier, static-embedding classifier, and lightweight contextual model such as DistilBERT.
- Report accuracy, macro-F1, weighted F1, per-class outcomes, confusion matrices, training time, and inference time.
- Analyse at least ten model disagreements, perform one ablation, and include one interpretability analysis.
- Conclude whether the contextual model's performance gain justifies its computational and interpretability trade-offs.

PS-5. Retrieval-Augmented Question Answering

Allocated groups: 5, 10, 15

Problem: A language model may generate fluent but unsupported answers. Retrieval grounding should provide relevant evidence and reduce unsupported claims.

Objective: Determine whether retrieval improves factual correctness and evidence support compared with answering without document grounding.

What the group must do:

- Use a narrow public collection of official documents or technical documentation, HotpotQA, SQuAD 2.0, or a synthetic knowledge base with gold evidence. Sources: https://hotpotqa.github.io/, https://rajpurkar.github.io/SQuAD-explorer/, and https://resources.data.gov/open-licenses/.
- Build a TF-IDF or BM25 keyword retriever, a dense retriever, and a small retrieval-grounded QA system; include a direct-answer baseline where feasible.
- Create at least 30 questions with expected answers and gold evidence. Report Recall@k or MRR, correctness, evidence support, completeness, citation faithfulness, and unsupported-claim rate.
- Analyse at least ten success, failure, or unsupported-answer cases and perform one retrieval ablation.
- Conclude which retrieval design best supports accurate, evidence-grounded answers.

4. Dataset and synthetic-data documentation

- For every dataset, record source URL or citation, access date, licence or terms, exact version, size, labels or tags, preprocessing, split procedure, and limitations.
- For synthetic data, record generator, templates or grammar, random seed, vocabulary, label rules, difficulty controls, noise process, train-validation-test construction, and examples. Generate final synthetic test items independently to prevent leakage.
- No sentence, document, template, question, or near-duplicate may occur in both training and final test sets.
- Do not claim real-world validity from synthetic-only results; discuss the difference between synthetic and naturally occurring language.

5. Safeguards and reproducibility

- Add a proposal page containing title, research question, dataset, baseline, advanced method, metrics, ablation, risks, and role allocation.
- For stochastic methods, use at least three seeds where computationally feasible and report mean and standard deviation; justify a single run when necessary.
- State exact pretrained model name and version, source, licence, download date, and whether it was frozen or fine-tuned.
- The instructor may conduct a short viva or demonstration in which any member explains the dataset, code, methods, and reported results.

6. Paper and submission requirements

- Paper length: 4,000–5,500 words excluding references and appendices. Required sections: Title; Abstract; Keywords; Introduction; Related Work; Dataset and Preprocessing; Methodology; Experimental Setup and Metrics; Results; Error Analysis and Discussion; Limitations, Ethics and Reproducibility; Conclusion; References; Appendix.
- Cite at least five scholarly sources in addition to dataset and tool documentation.
- Submission archive must contain research_paper.pdf, README.md, requirements.txt or environment.yml, source_code or notebooks, data or data_access_instructions.md, results, figures, dataset card, and contribution and AI-use statement.
- The README must explain installation, data or synthetic-data generation, execution order, expected runtime, hardware, random seed, and output locations.

7. Evaluation rubric — 20 marks

Criterion	Marks
Problem formulation and literature grounding	2
Dataset selection and documentation	2
Baseline and core implementation	4
Advanced method and technical depth	3
Experimental design and evaluation	3
Results, error analysis, and discussion	3
Research-paper quality	2
Reproducibility, contribution, and integrity	1
TOTAL	20

8. Final checklist

- The actual problem and research question are stated clearly.
- Dataset or synthetic-data generator is obtainable and documented.
- Baseline, advanced or comparison method, and ablation completed.
- Required metrics, figures, tables, and error analysis included.
- Code runs from the README; contribution and AI-use declarations included.
- One ZIP archive uploaded by the portal deadline.